

CSE 340 Computer Architecture

Spring 2023

Quiz 1

Marks: 15

ANSWER ALL THE QUESTIONS

Time: 30 mins

Name _____ ID _____

Section _____

C01

1. Suppose you are asked to choose the best performing CPU. [5]

Specification of Machine A -

Instruction count = 5×10^9 , Clock rate = 300MHz, CPI= 3.2

Specification of Machine B -

Instruction count = 3×10^9 , Clock rate = 400MHz, CPI= 3.8

- State which one is faster and by what factor.
- State how CPU time is affected by changes in Clock Rate and Instruction Count.

1. Suppose, we have the following Instruction count data from two different compilers running the same program, where the clock rate is 200MHz.

	Instruction Count for Each type			Number of Clock Cycles Needed		
	A	B	C	A	B	C
Compiler 1	3	1	2	1	2	3
Compiler 2	4	1	5	2	1	1

- Which compiler executes more instructions? (1)
- Which compiler has a lower CPI? (2)
- Which compiler is faster? (2)

C01

2. Consider the following classes of tasks for a particular CPU performing a task.

	IC (10^9)	CPI
Task A	3	2.6
Task B	1	3.7
Task C	5	6.3
Task D	1	3.1

- Calculate the average CPI [3]
- Suppose it has a clock rate of 1.8MHz, then what is the CPU time. [3]
- If the program runs on a reference PC with an execution time of 120 nanoseconds, calculate the SPECRatio. [2]
- State two differences between RISC AND CISC. [2]